Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
San Francisco, CA

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience building backend automation, cloud platforms, and AI-integrated applications using Python, Java, GCP, AWS, and Kubernetes. Current research in LLM-powered agent pipelines and orchestration.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, SQL, C/C++, Go.

AI & Orchestration: Gemini, Vertex AI, LangChain, MCP tool-calling, RAG pipelines, Vector Databases, PyTorch, TensorFlow, scikit-learn.

Cloud & DevOps: Google Cloud Platform, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, GitLab CI/CD, Jenkins, Azure, Linux, Bash.

Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs.

Databases: PostgreSQL, MySQL.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM agent pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Built provider-agnostic LLM abstraction layer (Gemini, GPT-4o, Llama, Qwen) enabling model routing across agents, demonstrating **25.7x latency reduction** (39.4s vs 1012.1s avg per run).

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3) in a regulated enterprise engineering environment, replacing legacy scripts and cutting deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services under strict change-control standards.
- Managed **HashiCorp Vault** secrets and IAM access policies across environments, and migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and Amazon EFS.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** on engineering best practices.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** cataloguing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).